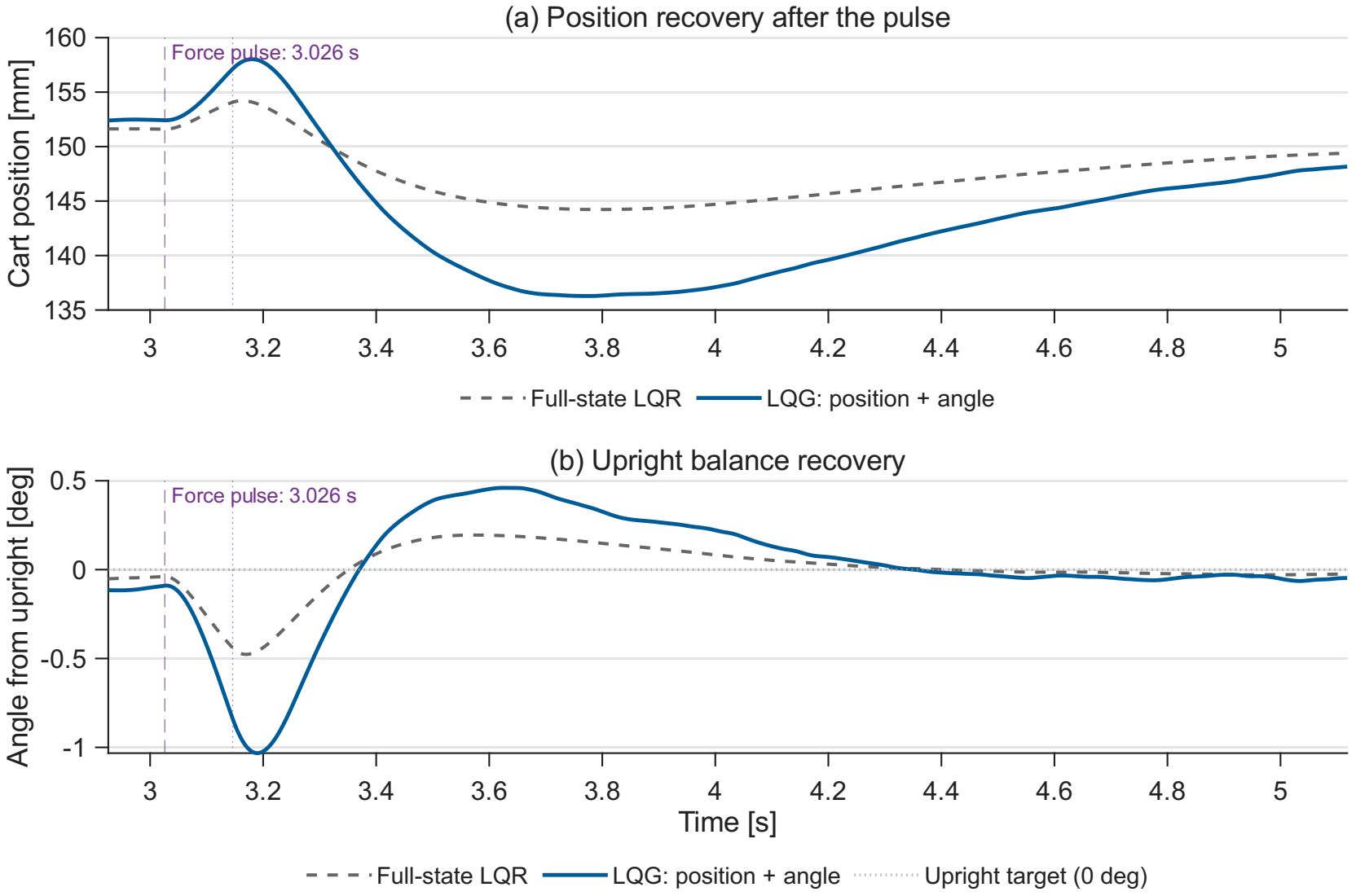

(a) Position recovery after the pulse
Force pulse: 3.026 s
Cart position [mm]
160
155
150
145
140
135
3
3.2
3.4
3.6
3.8
4
4.2
4.4
4.6
4.8
5
Full-state LQR
LQG: position + angle
(b) Upright balance recovery
Force pulse: 3.026 s
Angle from upright [deg]
0.5
0
-0.5
-1
Time [s]
Full-state LQR
LQG: position + angle
Upright target (0 deg)